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java. Assignment - 3

```
import java.util.InputMismatchException;  
import java.util.Scanner;
```

```
class InvalidMarksException extends Exception {  
    public InvalidMarksException (String message) {  
        super(message);  
    }  
}
```

```
class Student {  
    private int rollNumber;  
    private String studentName;  
    private int[] marks = new int[3];
```

```
    public Student (int rollNumber, String studentName, int[]  
        marks) throws InvalidMarksException {  
        this.rollNumber = rollNumber;  
        this.student = studentName;  
        validateMarks (marks);  
        this.marks = marks;  
    }
```

```
    private void validateMarks (int[] marks) throws InvalidMarksEx  
        ception {  
        for (int i = 0; i < marks.length; i++) {  
            if (marks[i] < 0 || marks[i] > 100) {  
                throw new InvalidMarksException ("Invalid marks" +  
                    marks[i]);  
            }  
        }  
    }  
}
```

```

public double CalculateAverage() {
    int total = 0;
    for (int marks : marks) {
        total += marks;
    }
    return total / 3.0;
}

```

```

public void display Result() {
    System.out.println("Roll Number" + rollNumber);
    System.out.println("Student Name" + student);
    System.out.println("S: " + marks);
    for (int m : marks) System.out.println(m + " ");
    System.out.println("In Average" + calculateAverage());
}

```

```

public int get RollNumber() {
    return rollNumber;
}

```

```

public class ResultManager {

```

```

    private Student[] students = new Student[100];
    private int studentCount = 0;
    private Scanner scanner = new Scanner(System.in);

```

```

    public void addStudent() {
        try {
            System.out.println("Enter Roll number");
            int roll = scanner.nextInt();
            scanner.nextLine();

```



```
System.out.println("Enter Student Name:");  
String name = Scanner.nextLine();
```

```
int[] marks = new int[3];  
for (int i = 0; i < 3; i++) {  
    System.out.print("Enter marks for subject " + (i+1) +  
        " : ");  
    marks[i] = Scanner.nextInt();  
}
```

```
Student [studentCount++] = new Student(roll, name,  
    marks);  
System.out.println("Student added!");
```

```
} catch (Exception e) {  
    System.out.println("Error: " + e.getMessage());  
    Scanner.nextLine();  
}
```

```
}  
  
public void ShowStudentDetails() {  
    System.out.print("Enter Roll Number:");  
    int roll = Scanner.nextInt();
```

```
    for (int i = 0; i < studentCount; i++) {  
        if (Student[i].getRollNumber() == roll) {  
            Student[i].displayResult();  
            return;  
        }  
    }
```

```
    System.out.println("No Student found");  
}
```

```

public void mainmenu () {
    while (true) {
        System.out.println ("1. Add Student");
        System.out.println ("2. Show Student Details");
        System.out.println ("3. Exit");
        System.out.print ("Enter Choice");

        int ch = Scanner.nextInt();
        switch (ch) {
            case 1 : addStudent(); break;
            case 2 : ShowStudentDetails(); break;
            case 3 : return;
            default : System.out.println ("Invalid");
        }
    }
}

```

```

public static void main (String[] args) {
    new ResultManager().mainmenu();
}

```